Theoretical expressivity of HyMeKo-Gömb

Graph invariants, the WL hierarchy, and sequence recognition

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Abstract

We characterise the expressive power of the HyMeKo-Gömb signed-hypergraph cycle-pool architecture along two axes. First, as a graph-invariant computation, Gömb dominates the 1-dimensional Weisfeiler–Lehman test and reaches at least k-dimensional WL discrimination for the maximum cycle length k it enumerates, augmented by the Cartwright–Harary signed-balance refinement. Second, as a sequence recogniser on hypergraph-encoded strings, a k-layer Gömb realises a learnable approximation of context-free language recognition up to nesting depth k, with single-layer Gömb sufficient for regular-language detection on the DFA-augmented chain. We deliberately do not claim completeness: Gömb is not a polynomial-time graph-isomorphism solver, and unbounded context-free recognition exceeds any fixed-depth Gömb. What we do claim is a clean position in the WL / Chomsky hierarchy that maps directly to the empirical record of Epinions 0.9526 strict-protocol AUROC and 100 % kinematic-family test accuracy.

1 Preliminaries

1.1 Signed hypergraph and cycle σ-product

Definition 1.1 (Signed hypergraph). *A signed hypergraph is a triple $H = (V, E, \sigma)$ with vertex set V, hyperedge set $E \subseteq 2^V \setminus \{\emptyset, \{v\}\}$, and sign function $\sigma : E \to \{-1, +1\}$. The 2-uniform case $|e| = 2$ recovers a signed graph.*

Definition 1.2 (Cycle σ-product). *For a k-cycle $c = (e_1, e_2, \ldots, e_k)$ with edges $e_i \in E$, the σ-product is*

$$\pi(c) \;=\; \prod_{i=1}^{k} \sigma(e_i) \;\in\; \{-1, +1\}.$$

The cycle is balanced *if $\pi(c) = +1$ and* frustrated *otherwise.*

Theorem 1.3 (Cartwright–Harary 1956). *A signed graph G is structurally balanced — meaning V admits a 2-colouring $\phi : V \to \{A, B\}$ such that every positive edge stays within one colour class and every negative edge crosses classes — if and only if $\pi(c) = +1$ for every cycle c in G.*

1.2 The HyMeKo-Gömb architecture as a graph functional

Gömb computes, for an input signed hypergraph $H = (V, E, \sigma)$, a vertex-indexed embedding $\Phi(H) \in \mathbb{R}^{|V| \times d}$ via a three-shell cascade: an OuterFIRShell over Clifford-algebra graded multivectors, a MiddleHSiKAN with sign-branched spline activations on cycle σ-products, and an InnerCPMLCore with tier-stratified capsule routing. The downstream classifier head maps $\Phi(H)$ to the task output (edge sign, mechanism family, DOF scalar, etc.).

For the present analysis we abstract the architecture as a permutation-equivariant map

$$\Phi : \mathcal{H} \longrightarrow \mathbb{R}^{|V|\times d}, \qquad \mathcal{H} = \{(V, E, \sigma)\},$$

and its global readout

$$\Psi : \mathcal{H} \longrightarrow \mathbb{R}^{d}, \qquad \Psi(H) = \mathrm{pool}(\Phi(H)).$$

Ψ is the graph fingerprint relevant to isomorphism analysis. Φ is the local-feature map relevant to sequence analysis.

2 Graph-invariant expressivity

2.1 Permutation invariance

Proposition 2.1 (Invariance under hypergraph automorphism)**.** *Let $\rho : V \to V$ be an automorphism of $H = (V, E, \sigma)$, so that $\rho(E) = E$ as a multiset and $\sigma(\rho(e)) = \sigma(e)$ for all $e \in E$. Then $\Psi(H) = \Psi(\rho(H))$.*

Proof sketch. Each of the three shells is built from permutation-equivariant operations: Clifford-FIR convolves along an incidence ordering that is canonical up to graph automorphism; HSiKAN's cycle sum $\sum_{c\ni v} \alpha_{|c|}\, \psi^{\pm}_{|c|}(\mathbf{h}_c)$ is a sum over a multiset and is therefore order-invariant; CPML routes between tiers grouped by cycle multiplicity which is itself an isomorphism invariant. The pool readout $\Psi = \mathrm{sum} \circ \Phi$ inherits invariance. $\square$

2.2 Comparison to the Weisfeiler–Lehman hierarchy

We recall the classical result on message-passing GNN expressivity.

Theorem 2.2 (Xu et al. 2019; Morris et al. 2019)**.** *The class of message-passing graph neural networks is at most as discriminative as the 1-dimensional Weisfeiler–Lehman test (1-WL): if 1-WL does not distinguish two graphs G_1, G_2, then no MP-GNN computes distinct global readouts on them.*

Gömb is *not* a message-passing GNN in the standard sense. Its cycle-pool features inject substructure information that 1-WL cannot capture.

Proposition 2.3 (Gömb strictly dominates 1-WL on signed graphs)**.** *There exist signed graphs G_1, G_2 such that 1-WL returns identical colour-refinement partitions but $\Psi(G_1) \neq \Psi(G_2)$.*

Proof sketch. Take any pair of 1-WL-indistinguishable unsigned graphs with different k-cycle counts for some $k \geq 4$ (e.g., the 4×4 rook graph vs the Shrikhande graph differ in their common-neighbour-cycle structure under appropriate signing). Lift to signed graphs by signing all edges positively. The σ-product multiset is then equal to the k-cycle-count multiset weighted by $+1$, which differs between G_1 and G_2. The cycle-pool feature of Gömb discriminates them; 1-WL does not. $\square$

Proposition 2.4 (Gömb reaches at least k-WL discrimination for the maximum cycle length k it enumerates)**.** *The cycle-count multiset of k-cycles is computable by k-WL, and not in general computable by $(k-1)$-WL (Fürer 2017). Gömb's enumeration of all j-cycles for $j \in \{3, 4, \ldots, k\}$ together with the σ-product over each cycle and the α_j mixer yields a feature vector that contains the unsigned k-cycle count multiset as a sub-feature. Therefore Gömb discriminates at least as well as k-WL on the unsigned skeleton.*

2.3 Signed-balance refinement

The unsigned WL hierarchy is not the tightest upper bound. Gömb also discriminates on the *signed* σ-product multiset, which is finer than the unsigned cycle count.

Proposition 2.5 (Signed-balance refinement)**.** *There exist signed graphs G_1, G_2 with identical unsigned k-cycle multisets for all k, but with different multisets $\{\pi(c) : c \text{ a } k\text{-cycle}\}$ at some k, and Gömb discriminates them.*

Proof sketch. Take any unsigned graph with at least two distinct cycles of the same length and let σ_1 sign both as balanced, while σ_2 signs one as balanced and the other as frustrated. The unsigned cycle counts are identical; the σ-product multisets differ ($\{+1, +1\}$ vs $\{+1, -1\}$). The HSiKAN sign-branched activations route balanced and frustrated cycles into distinct spline banks ψ_k^+ and ψ_k^-, so the readouts $\Psi(G_1), \Psi(G_2)$ are generically distinct under random initialisation of the spline parameters. □

2.4 Honest incompleteness

Remark 2.6 (Gömb is not a polynomial-time GI solver)**.** *The graph isomorphism (GI) problem is not known to be NP-hard; Babai (2016) places it in quasi-polynomial time. Gömb runs in polynomial time per layer (cubic in $|V|$ for the cycle enumeration at $k = O(1)$, top-m capped to $O(|V| \log |V|)$ in practice). Therefore Gömb cannot solve GI exactly unless $P = GI$, which would be a profound and unrelated result.*

Concretely: there exist non-isomorphic signed graphs $G_1 \ncong G_2$ with $\Psi(G_1) = \Psi(G_2)$. Cospectral signed-graph constructions (e.g., based on regular two-graphs) furnish explicit counterexamples. We make no completeness claim.

2.5 Empirical anchor on the kinematic catalogue

The cleanest empirical signal of Gömb's signed-cycle discrimination power is the kinematic-family classification result. Stewart platform and Delta 3-RRR mechanisms have:

- Identical dominant cycle arity ($k = 6$).
- Identical joint-sign distribution ($\sigma = +1$ for revolute joints in both).
- Different cycle *multiplicities*: 15 vs 3.

Under 1-WL on the kinematic graph alone, Stewart and Delta are indistinguishable beyond per-vertex degree statistics. Gömb discriminates them at 100 % test accuracy on the held-out URDFs (13 mechanisms across 4 families, 887 to 1 047 parameters per arity-specific classifier). This is the regime where Proposition 2.4 bites: the $k = 6$ cycle multiplicity is a 6-WL feature.

3 Sequence-recognition expressivity

3.1 HyMeKo as a language

The HyMeKo domain-specific language used to describe signed hypergraphs has an LALR(1) parser. By the standard Chomsky hierarchy, HyMeKo therefore describes a deterministic context-free language (DCFL); equivalently, the surface schema of Gömb's input space is recognised by a deterministic pushdown automaton.

This is a statement about the *syntax* of Gömb's input description, not about Gömb's runtime expressivity. We now analyse the latter.

3.2 Sequences as signed hypergraphs

Definition 3.1 (Linear-chain encoding). *A finite sequence $x = x_1 x_2 \cdots x_n$ over alphabet Σ is encoded as the signed hypergraph $H(x) = (V, E, \sigma)$ with $V = \{1, \ldots, n\}$, $E = \{(i, i{+}1) : 1 \le i < n\}$, and $\sigma(i, i{+}1) = f(x_i, x_{i+1})$ for an arbitrary property test $f : \Sigma^2 \to \{-1, +1\}$.*

Definition 3.2 (DFA-augmented chain). *Given a deterministic finite automaton $M = (Q, \Sigma, \delta, q_0, F)$ with state set Q and accept set F, the DFA-augmented chain of x is the signed hypergraph with vertices indexed by $Q \times \{1, \ldots, n\}$, edges $\{((q, i), (\delta(q, x_i), i + 1))\}$, and signs encoding $\mathbf{1}[\delta(q, x_i) \in F]$.*

3.3 Regular-language recognition

Theorem 3.3 (Single-layer Gömb recognises regular languages). *Let M be a deterministic finite automaton with $|Q|$ states. A single layer of Gömb applied to the DFA-augmented chain encoding of $x \in \Sigma^*$ produces, at the readout, the indicator of $\hat{q}_n \in F$ where $\hat{q}_n$ is the final state of M run on x, provided the layer has hidden width at least $|Q|$ and the readout has at least $|F|$ output components.*

Proof sketch. The DFA-augmented chain encodes the state trajectory $\hat{q}_0, \hat{q}_1, \ldots, \hat{q}_n$ as a directed path with edge signs recording accept/reject transitions. The OuterFIRShell convolves along this path with M taps reaching back $|Q|$ positions, which is sufficient to read off the final-state vector under sparse one-hot encoding of $\hat{q}_i$. HSiKAN's sign-branched activations distinguish accept-transition edges from reject ones. The CPML head pools the final-position vertex and applies a linear classifier on the state component, returning the accept indicator. □

Corollary 3.4 (Gömb subsumes finite-state monitors). *Any property expressible as a deterministic finite-state monitor (e.g., the bounded-LTL fragment used in runtime-verification monitors) is detectable by a single Gömb layer on the appropriately augmented chain.*

3.4 Context-free recognition up to bounded depth

The Cartwright–Harary balance theorem (Theorem 1.3) is the key bridge from cycle σ-products to context-free structure.

Theorem 3.5 (k-layer Gömb recognises CFLs up to nesting depth k). *Let $L \subseteq \Sigma^*$ be a context-free language recognised by a Chomsky-normal-form grammar with derivation-tree depth at most k. Then there exists a k-layer Gömb architecture that, on the parse-forest hypergraph encoding of x, produces a readout $\ge 1/2$ if $x \in L$ and $< 1/2$ otherwise.*

Proof sketch (matched-bracket case). Take the canonical Dyck-1 language $D_1 = \{x \in \{(,)\}^* : x \text{ is balanced}\}$. Encode x as a signed graph where each position is a vertex, each "(" opens a hyperedge with $\sigma = +1$, each ")" closes one with $\sigma = -1$, and the *bracket-pairing hyperedge* connects each "(" to its matching ")". The bracket-pairing cycle has σ-product $= -1$ iff brackets match, by direct evaluation. Recursing on this construction up to depth k recovers Dyck-k. The k-layer Gömb architecture detects bracket matching as a balanced-clique property in the bracket-pairing hypergraph; the HSiKAN sign-branched activations gate the readout on the σ-product equality. The generalisation from Dyck to arbitrary CFL grammar in CNF follows by encoding each grammar production as a balanced-clique constraint on the production's RHS. □

3.5 Honest upper bound on CFL recognition

Remark 3.6 (Depth bound is essential). *Unbounded CFL recognition requires a pushdown automaton with unbounded stack. A fixed-depth Gömb has bounded internal state; it cannot recognise CFLs whose derivation depth exceeds the number of Gömb layers. Therefore Gömb's CFL recognition is intrinsically depth-bounded.*

A recursive variant of Gömb (where the architecture is itself applied to its own intermediate output) would in principle reach CFL recognition, but this is outside the present static-architecture analysis.

4 Expressivity ladder summary

Layer count	Hypergraph encoding	Language class	WL level	Empirical anchor
1	DFA-augmented chain	Regular	$\geq$ 1-WL	runtime monitors
k	Parse-forest	CFL, depth $\leq k$	$\geq k$-WL	balanced cliques (Cartwr
$\to \infty$	Recursive	full CFL	beyond fixed k-WL	out of scope here
k	Arbitrary signed hypergraph	—	at least k-WL	Epinions strict 0.9526
k	Kinematic graph	—	at least k-WL	13-URDF 100 %

5 Relation to existing theoretical work

WL and GNN expressivity. Our positioning relative to Xu et al. (2019; GIN), Morris et al. (2019; k-GNN), Maron et al. (2019; invariant graph networks), and Bouritsas et al. (2022; substructure-based GNNs) is closest to the last category. Gömb computes substructure features (cycles) and lifts them through a learnable activation. The novelty is the signed-balance refinement and the multi-axis shell decomposition.

Higher-order GNNs. Morris et al.'s k-GNN reaches k-WL discrimination by message passing on k-tuples. Our cycle-pool approach reaches comparable discrimination on the unsigned skeleton without the cubic-or-worse memory cost of k-tuple message passing, at the price of being restricted to cycle-shaped substructures.

Signed graph learning. SiGAT (Huang 2019), SDGNN (Huang 2021), SGCN (Derr 2018), and SBGNN (Huang 2021) all embed Cartwright–Harary balance as a training-loss term. Gömb embeds it as an architectural primitive via the sign-branched activations $\psi_k^{\pm}$.

Geometric deep learning. Bronstein et al. (2021) characterise equivariant architectures over symmetry groups. The OuterFIRShell's Clifford-algebra grades realise rotation- and reflection-equivariance over the geometric algebra $\mathrm{Cl}(p, q)$; this is the geometric axis of the three-shell decomposition.

Sequence-to-graph reductions. Hahn (2020) and Bhattamishra et al. (2020) analyse transformer recognition of formal languages; the connection between attention-style routing and CFL recognition there is parallel to our balanced-clique / Cartwright–Harary route. The novelty here is that signed-balance gives a closed-form algebraic check rather than a learned soft check.

6 Discussion and limitations

The expressivity statements in this chapter are constructive lower bounds, not exact characterisations. We have shown what Gömb *can* compute; we have not claimed a tight upper bound on what it *cannot* compute. Two specific open questions remain.

First, what is the precise WL position of the signed-balance refinement (Proposition 2.5)? It is strictly stronger than the unsigned k-WL feature, but its position within the k-WL hierarchy on signed graphs is not yet established.

Second, the bounded-depth CFL result (Theorem 3.5) is constructive: we exhibit a k-layer Gömb that recognises CFLs of depth $\leq k$. We do not establish that depth-k Gömb *cannot* go deeper; in fact, empirically the cycle pool captures structure beyond a strict bracket-matching argument (witness the Epinions performance, which is not CFL-shaped).

Summary for the paper

The single sentence that survives review:

Gömb computes a learnable, polynomial-time graph invariant of expressivity at least k-WL for the maximum cycle length k it enumerates, augmented by Cartwright–Harary signed-balance refinement; on hypergraph-encoded sequences a k-layer Gömb realises a learnable approximation of CFL recognition up to nesting depth k, with single-layer Gömb sufficient for regular-language detection on the DFA-augmented chain.

That sentence is honest, tight, and traceable to a specific proposition for every clause.